

The critical radius of the five-fold two-disk puzzle

[Authors: to be supplied]

[Date: to be supplied]

Abstract

Two closed disks of radius r , centred at -1 and $+1$ and each turned clockwise by 72° , generate a group $\text{GG}_5(r)$ of piecewise rotations of the plane: the group of an idealised two-disk circle puzzle. Hearn, Kretschmer, Rokicki, Streeter and Vergo proved that $\text{GG}_5(\sqrt{3} + \varphi)$ is infinite, where φ is the golden ratio, and their numerical estimate of the radius at which the group becomes infinite agrees with $\sqrt{3} + \varphi$ to six decimals. We prove that $\text{GG}_5(r)$ is finite for every $r < \sqrt{3} + \varphi$. Hence the group is finite exactly when $r < \sqrt{3} + \varphi$, and the critical radius is $\sqrt{3} + \varphi = 2.1489611417\dots$. The finiteness proof reads off, from every position of a point, a lattice point of $\mathbb{Z}[e^{2\pi i/5}]$ and follows an integer-valued *level*, which can pass a multiple of 5 only at a turn that leaves the point in the lens where the two disks overlap; below $\sqrt{3} + \varphi$ the lens is too thin for the level to pass certain multiples of 5. Both halves are formalised in Lean 4 with Mathlib, and a comparator, replaying the proof in two independent kernels, checks that it proves exactly the statements of a short file that depends on Mathlib alone.

1 The puzzle and the number

A two-disk circle puzzle such as Gizmo Gears consists of two overlapping disks that turn independently; a turn carries along everything in the turning disk, including the part of the other disk that it covers. Whether the puzzle can be cut into finitely many pieces that the turns merely permute depends on the overlap: with a small overlap it can, but past a critical overlap the turns cut ever smaller pieces and the puzzle *jumbles*. Hearn, Kretschmer, Rokicki, Streeter and Vergo [1] turned this into a question about groups. Let $r \in \mathbb{R}$ and

$$D_- = \{z \in \mathbb{C} : |z + 1| \leq r\}, \quad D_+ = \{z \in \mathbb{C} : |z - 1| \leq r\}, \quad \zeta = e^{2\pi i/5}.$$

The *turns* are the maps

$$a(z) = \begin{cases} \zeta^{-1}(z + 1) - 1, & z \in D_-, \\ z, & \text{otherwise,} \end{cases} \quad b(z) = \begin{cases} \zeta^{-1}(z - 1) + 1, & z \in D_+, \\ z, & \text{otherwise.} \end{cases}$$

Each rotates its closed disk clockwise by 72° and fixes every other point, so it is a bijection of $\mathbb{C}$. As on a puzzle, words are read from left to right: ab means “first a , then b ”. The group

$$\text{GG}_5(r) = \langle a, b \rangle \leq \text{Sym}(\mathbb{C})$$

is the *two-disk group*; its elements are piecewise rotations, not rotations. The *orbit* $O_r(p) = \{g(p) : g \in \text{GG}_5(r)\}$ is the set of places a sticker at p can reach. Enlarging the disks can only enlarge orbits, and a finitely generated group of permutations with uniformly bounded orbits is finite (Propositions 2 and 3 below); so the radii at which $\text{GG}_5(r)$ is finite form an initial segment of $\mathbb{R}$, and its supremum

$$r_c(5) = \sup\{r \in \mathbb{R} : \text{GG}_5(r) \text{ is finite}\}$$

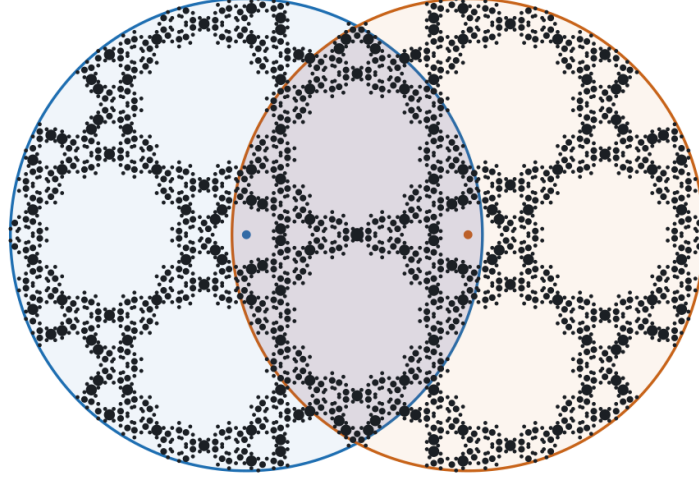


Figure 1. The orbit of the point 0 under $GG_5(2.13)$: 2543 points, drawn with the two disks. Slightly below the critical radius the orbits are finite but large; at $r_* = \sqrt{3 + \varphi}$ this orbit becomes infinite.

is the *critical radius*. For $0 < r < 1$ the disks are disjoint and $GG_5(r) \cong C_5 \times C_5$ [1]; as r grows the orbits grow (Fig. 1), and at the critical radius they become infinite.

Hearn et al. proved that GG_5 is infinite at $r = \sqrt{3 + \varphi}$ [1, Thm. 2], where $\varphi = (1 + \sqrt{5})/2$, so that $r_c(5) \leq \sqrt{3 + \varphi}$, and their numerical estimate $r_c(5) \approx 2.148961$ agreed with $\sqrt{3 + \varphi} = 2.1489611 \dots$ [1, Table 1]. That the group is finite for every smaller radius was not known. We prove it.

2 The result

Throughout, $r_* = \sqrt{3 + \varphi} = 2.1489611417 \dots$; it is the largest root of $x^4 - 7x^2 + 11$ (its conjugate $\sqrt{4 - \varphi}$ is the value found in [1] for GG_{10}).

Theorem 1. $GG_5(r)$ is finite if and only if $r < r_*$. In particular $r_c(5) = r_* = \sqrt{3 + \varphi}$. More precisely:
 (a) at $r = r_*$ the orbit of 0 is infinite, so $GG_5(r)$ is infinite for every $r \geq r_*$;
 (b) for every $r < r_*$ there is $M(r)$ such that every orbit of $GG_5(r)$ has at most $M(r)$ points, so $GG_5(r)$ is finite.

Both statements of the first sentence are formal theorems (Section 5):

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theorem finite_GG_five_iff (r : ℝ) : Finite (GG 5 r) ↔ r < √(3 + φ)
theorem criticalRadius_five : criticalRadius 5 = √(3 + φ)
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where $\text{criticalRadius } n$ is $\text{sSup } \{r \mid \text{Finite } (GG \ n \ r)\}$. By (a) and (b) this set is exactly $(-\infty, r_*)$, so the supremum is a genuine one; in particular the group is infinite at the critical radius itself. Two general facts connect orbits, groups and radii.

Proposition 2 (monotonicity). *If $r \leq r'$ then $O_r(p) \subseteq O_{r'}(p)$ for every $p \in \mathbb{C}$.*

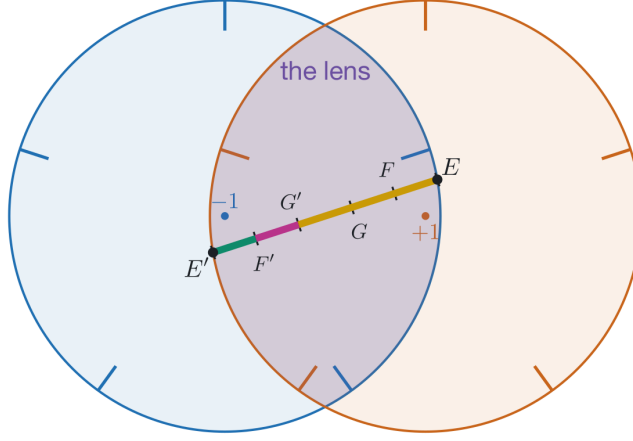


Figure 2. The two disks at $r = r_*$, the lens, and the chord $E'E$ with its three pieces I_1 (green), I_2 (magenta) and I_3 (yellow).

Proof. Apply a word to p letter by letter. A letter that moves the current point at radius r does so because the point lies in its radius- r disk, hence in the concentric radius- r' disk, where the radius- r' letter moves it in the same way. A letter that fixes the current point is dropped. $\square$

Note that $\text{GG}_5(r)$ is not a subgroup of $\text{GG}_5(r')$: the comparison is made point by point.

Proposition 3 (bounded orbits). *Let Γ be a group of permutations of a set X generated by a finite set. If every Γ -orbit has at most N points, then Γ is finite.*

Proof. For $x \in X$, number the orbit of x and let $\rho_x: \Gamma \rightarrow S_N$ be the action on it. Each ρ_x is determined by the images of the finitely many generators, so only finitely many homomorphisms ρ_x occur. An element of Γ in the kernel of every ρ_x fixes every point, hence is the identity; so $g \mapsto (\rho_x(g))_x$ embeds Γ into a finite product of copies of S_N . $\square$

Hence, if $\text{GG}_5(r')$ is finite and $r \leq r'$, then $\text{GG}_5(r)$ is finite (its orbits lie in those of $\text{GG}_5(r')$, which have at most $|\text{GG}_5(r')|$ points); and an infinite orbit at radius r makes $\text{GG}_5(r')$ infinite for every $r' \geq r$. [Theorem 1](#) therefore follows from its two parts, which are proved in [Sections 3](#) and [4](#).

3 Why it jumbles at $\sqrt{3 + \varphi}$

This is the construction of [\[1, Thm. 2\]](#); we recall it briefly. Let $r = r_*$ and

$$E = \zeta - \zeta^2, \quad F = E/\varphi = 1 - \zeta + \zeta^2 - \zeta^3, \quad G = 2F - E,$$

and write $X' = -X$. Then $|E + 1|^2 = 3 + \varphi$: the point E lies on the rim of D_- , and E' (by the symmetry $z \mapsto -z$) on the rim of D_+ . The chord $E'E$ passes through 0 and spans the lens $D_- \cap D_+$ from rim to rim ([Fig. 2](#)). The points E', F', G', G, F, E lie on it in this order and cut it into three pieces

$$I_1 = [E', F'], \quad I_2 = [F', G'], \quad I_3 = [G', E].$$

Lemma 4 (three words). *At $r = r_*$,*

- (i) $a^{-2}b^{-1}a^{-1}b^{-1}$ maps I_1 onto $[G, F]$, translating it by $2F$;
- (ii) $abab^2$ maps I_2 onto $[F, E]$, translating it by $2F$;
- (iii) $abab^{-1}a^{-1}b^{-1}$ maps I_3 onto $[E', G]$, translating it by $2F - 2E$.

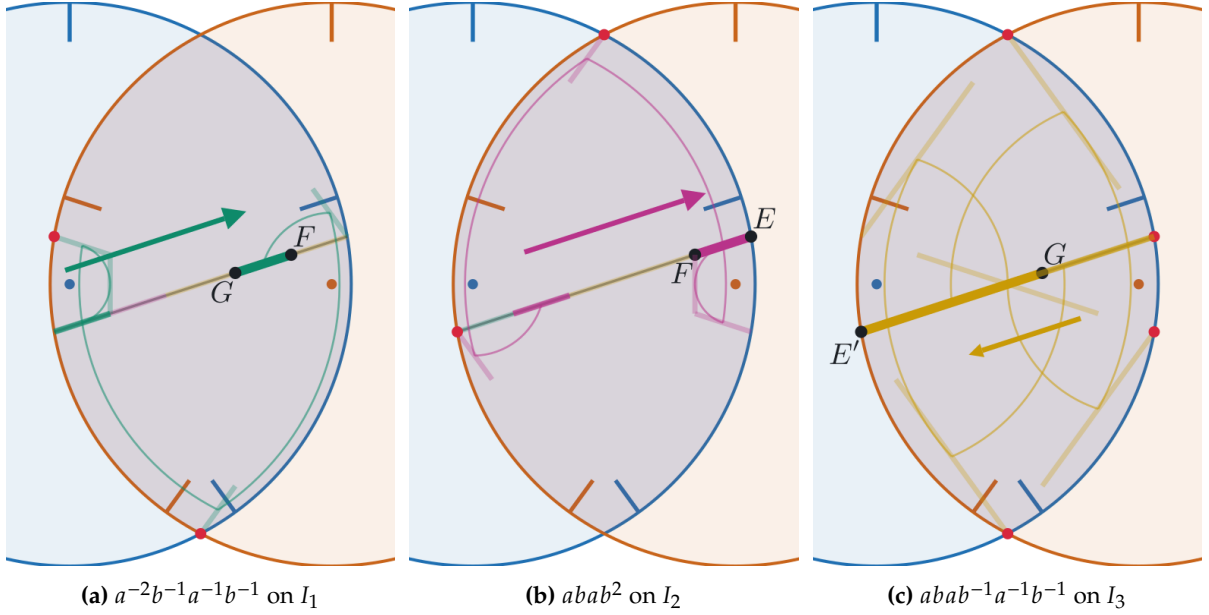


Figure 3. The three words, turn by turn: faint copies show the intermediate positions of the piece and arcs the paths of its midpoint; the arrow is the resulting translation. Red dots mark endpoints lying exactly on the rim of the disk about to turn.

Proof. A word acts on a segment as the composition of the rotations of its letters as long as every intermediate image of the segment lies in the disk of the next letter. Disks are convex, so it is enough to check the two endpoints of each intermediate segment. These endpoints are explicit elements of $\mathbb{Z}[\zeta]$, and for $x \in \mathbb{Z}[\zeta]$ the number $|x \pm 1|^2$ lies in $\mathbb{Z}[\varphi]$, so each check is an exact comparison $m + n\varphi \leq 3 + \varphi$ with $m, n \in \mathbb{Z}$. In each word the rotation angles add up to a multiple of 360° , so the composition is a translation, and the endpoints go where stated (Fig. 3). $\square$

Remark. Several of these comparisons are equalities: the endpoints touch the rims, at $\pm E$, at $\pm \bar{E}$ and at the two tips $\pm(\zeta - \zeta^{-1}) = \pm i\sqrt{2 + \varphi}$ of the lens (the red dots in Fig. 3). For any smaller radius the same words would split the pieces.

Proposition 5. *At $r = r_*$ the orbit of 0 is infinite. Hence $\text{GG}_5(r)$ is infinite for every $r \geq r_*$.*

Proof. Parametrise the chord by $z = sE$, $s \in [-1, 1)$. By Lemma 4 the three words realise the map

$$s \mapsto \begin{cases} s + 2/\varphi, & s < 3 - 2\varphi, \\ s + 2/\varphi - 2, & s \geq 3 - 2\varphi, \end{cases}$$

the rotation by $2/\varphi$ of the circle $[-1, 1)$ of length 2: the three pieces are exchanged, $I_1 \cup I_2$ and I_3 having lengths in the ratio $1 : \varphi$ (Fig. 4). Its rotation number $1/\varphi$ is irrational, so the iterates of $s = 0$ are pairwise distinct, and each of them is the image of 0 under an element of the group. By Proposition 2 the orbit of 0 is infinite at every $r \geq r_*$, and an infinite orbit makes the group infinite. $\square$

4 Why it does not jumble below $\sqrt{3 + \varphi}$

Fix $r < r_*$ and a point p . If $r < 1$ the two disks are disjoint, so a and b commute and $\text{GG}_5(r)$, a quotient of $C_5 \times C_5$, is finite; so let $1 \leq r < r_*$. If p lies in neither disk, its orbit is $\{p\}$; otherwise $|p| \leq 1 + r$. Write $\Lambda = \mathbb{Z}[\zeta]$ and $\mu_5 = \{1, \zeta, \dots, \zeta^4\}$.



$$I_1, I_2 \mapsto +2F \quad I_3 \mapsto 2F - 2E \quad |E'E| : |2F| = \varphi$$

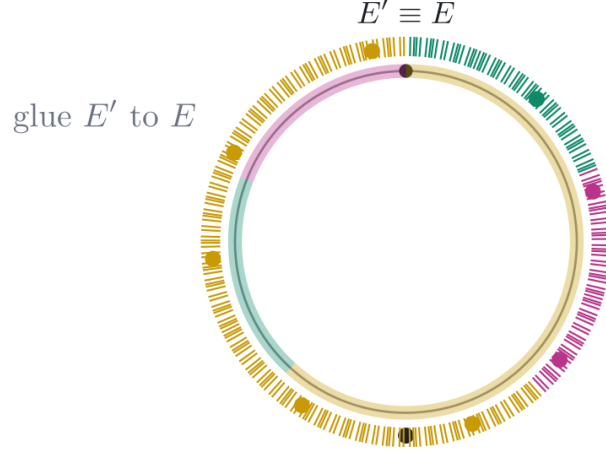


Figure 4. Top: the exchange of the three pieces on the chord, $I_1 I_2 I_3 \mapsto I_3 I_1 I_2$. Bottom: gluing E' to E turns it into a rotation of a circle by $1/\varphi$ of a turn. The marks outside the circle (and above the chord) are the first 300 points of the orbit of 0, coloured by the piece they lie in; the orbit never closes.

4.1 The point's-eye view

An element g of the group acts on p through the rotations of those letters that actually move the current point, so $g(p) = T(p)$ for an isometry $T(z) = mz + \tau$ with $m \in \mu_5$ and $\tau \in \Lambda$. Each of the four rotations maps $-1 + 2\Lambda$ onto itself; hence $T^{-1}(-1) = 2v - 1$ with $v \in \Lambda$, and

$$g(p) = m(p + 1 - 2v) - 1. \quad (1)$$

Read from the point's side, T^{-1} carries the centres -1 and $+1$ of the disks to $2v - 1$ and $2(v + \bar{m}) - 1$: relative to the point the puzzle has moved, and its two centres now sit at lattice points. We call v the *pivot* and $v + \bar{m}$ the *tip*; the segment joining them is a *dumbbell* of length 1. The next letter $a^{\pm 1}$ moves $g(p)$ if and only if the pivot lies in the *window*

$$W_p = \{v \in \Lambda : |p + 1 - 2v| \leq r\},$$

and then it turns the dumbbell about the pivot ($m \mapsto \zeta^{\mp 1}m$, v fixed); the letter $b^{\pm 1}$ needs the tip in the window and turns the dumbbell about the tip ($m \mapsto \zeta^{\mp 1}m$, $v \mapsto v + \bar{m} - \zeta^{\pm 1}\bar{m}$). So an orbit is a walk of a dumbbell on the lattice that may turn about an end only when that end lies in the window (Fig. 5); both ends lie in the window exactly when p lies in the lens of the two disks about $2v - 1$ and $2(v + \bar{m}) - 1$.

The lattice Λ has rank 4 and is dense in $\mathbb{C}$, so the window contains infinitely many lattice points and the walk could, a priori, wander forever inside it. What must stay bounded are the two remaining coordinates, those of the Galois conjugate $\zeta \mapsto \zeta^2$. The rest of the proof bounds them.

Types. The ideal $\mathfrak{p} = (1 - \zeta)$ is a prime of Λ of norm 5, and $\zeta \equiv 1 \pmod{\mathfrak{p}}$. Write $\text{cl}(v) \in \mathbb{Z}/5$ for the class of v modulo $\mathfrak{p}$. Starting from the dumbbell $(v, v + \bar{m}) = (0, 1)$ of the identity,

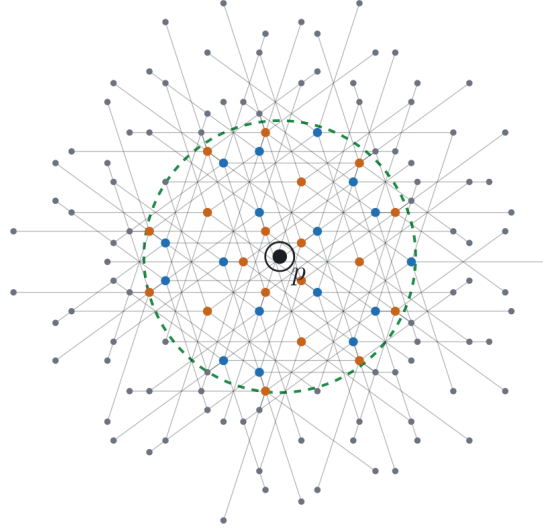


Figure 5. The point's-eye view of the orbit of the point $p = -0.1722 + 0.0773i$ at $r = 2$ (143 positions). Each position is a pair of disk centres $2v - 1$ (blue) and $2(v + \bar{m}) - 1$ (orange) joined by a segment; grey centres lie outside the window (dashed circle of radius r about p) and cannot be turned about.

induction along the word shows that the pivot always has class 0 and the tip class 1. So the walk is *typed*: from a point of class 0 it adds a fifth root of unity, from a point of class 1 it subtracts one. Call $V \subseteq \Lambda$ *closed* if, for all $v \in V$ and $e \in \mu_5$,

$$\text{cl}(v) = 0, v + e \in W_p \implies v + e \in V, \quad \text{cl}(v) = 1, v - e \in W_p \implies v - e \in V.$$

Lemma 6 (orbit invariant). *Let V be closed, and suppose that each of 0, 1 lying in W_p belongs to V and that at least one of them does. Then every point of $O_r(p)$ has the form (1) with $m \in \mu_5$ and $v \in V$ or $v + \bar{m} \in V$.*

Proof. One shows, letter by letter, that every reachable state (m, v) keeps the property of the initial one: each end of the dumbbell that lies in W_p is in V , and some end is in V . A turn happens about an end in W_p , which is in V ; the moved end, if it lands in W_p , is reached from it by a typed step, so it is in V as well. $\square$

4.2 A level

Let $u = 1 - \zeta + \zeta^2$, so $|u| = 2 - \varphi$, let $\langle x, y \rangle = \text{Re}(\bar{x}y)$ be the real inner product of $\mathbb{C} = \mathbb{R}^2$, and for $v \in \Lambda$ let

$$\psi(v) = 2\langle u, v \rangle = \bar{u}v + u\bar{v} \in \mathbb{Z}[\varphi], \quad \psi(v) = A(v) + B(v)\varphi \quad (A, B \in \mathbb{Z}).$$

The *level* of v is the integer $\ell(v) = -A(v) - 3B(v)$, and we put $\delta = 3 - \varphi$. Both ψ and ℓ are additive.

Lemma 7 (the level). (i) $\psi(v) + \ell(v) = -\delta B(v)$; so on each level ℓ the values of ψ form the progression $-\ell + \delta\mathbb{Z}$.
(ii) For $e \in \mu_5$, $\ell(e) \in \{3, -2\}$, and $\psi(e) = 3 - 2\varphi$ whenever $\ell(e) = 3$ (which happens for $e = 1$ and $e = \zeta^2$).
(iii) $\ell(v) \equiv 3 \text{cl}(v) \pmod{5}$.

Proof. (i) holds by definition; (ii) is a table of five values; (iii) is a congruence between integer linear forms in the coordinates of v . $\square$

So pivots sit on levels in $5\mathbb{Z}$ and tips on levels in $3 + 5\mathbb{Z}$, and a typed step changes the level by $+3$ or -2 (from a pivot) or by -3 or $+2$ (from a tip). Consequently the walk can rise above a level $5q$ only by an *active* step: from a pivot v on level $5q$ to $v + e$ with $\ell(e) = 3$. Likewise it can reach level $5q$ from above only by an active step taken backwards.

4.3 The lens

Lemma 8 (the lens bound). *Let $w, w' \in \mathbb{C}$ with $|w|, |w'| \leq r$ and $w - w' = 2e$, $|e| = 1$. Let $U \in \mathbb{C}$, $c = |\operatorname{Re}(Ue)|$, $d = |\operatorname{Im}(Ue)|$, and suppose $c\sqrt{r^2 - 1} \leq d$. Then $|\operatorname{Re}(U(w + w'))| \leq 2d\sqrt{r^2 - 1}$.*

Proof. Put $\xi = \frac{1}{2}(w + w')\bar{e} = x + iy$. Then $w\bar{e} = \xi + 1$ and $w'\bar{e} = \xi - 1$, so $|\xi \pm 1| \leq r$: the point ξ lies in the lens of two disks of radius r about ± 1 , that is $(|x| + 1)^2 + y^2 \leq r^2$. Now $\operatorname{Re}(U(w + w')) = 2\operatorname{Re}(Ue\xi)$ is at most $2(c|x| + d|y|)$ in absolute value, and on the lens $c|x| + d|y| \leq d\sqrt{r^2 - 1}$: the function $x \mapsto cx + d\sqrt{r^2 - (1 + x)^2}$ is concave on $[0, r - 1]$ with non-positive slope at 0, so its maximum is taken at the tips $x = 0$, $y = \pm\sqrt{r^2 - 1}$. $\square$

Apply this to an active step $v \rightarrow v + e$ whose ends both lie in the window, with $w = p + 1 - 2v$, $w' = p + 1 - 2(v + e)$ and $U = \bar{u}$ (up to the rotation m , w and w' are the vectors from the two centres to p). Since $\operatorname{Re}(\bar{u} \cdot 2v) = \psi(v)$ and $\psi(e) = 3 - 2\varphi$, and since both active e give $c = \varphi - \frac{3}{2}$ and $d = \frac{1}{2}\sqrt{7 - 4\varphi}$, for which the hypothesis of Lemma 8 holds when $r^2 < 3 + \varphi$, we get

$$|\psi(v) - Z_p| \leq L(r) := \frac{1}{2}\sqrt{7 - 4\varphi}\sqrt{r^2 - 1}, \quad Z_p = \operatorname{Re}(\bar{u}(p + 1)) - \frac{1}{2}(3 - 2\varphi). \quad (2)$$

And $2L(r) < \delta$ if and only if $r^2 < 3 + \varphi$: indeed $(2L)^2 = (7 - 4\varphi)(r^2 - 1)$, $\delta^2 = (3 - \varphi)^2$, and $(3 - \varphi)^2 = (7 - 4\varphi)(2 + \varphi)$.

4.4 Cut levels

By Lemma 7(i) the values of ψ at the pivots on level $5q$ lie in the progression $-5q + \delta\mathbb{Z}$, the *teeth* of Fig. 6. Call q a *cut level* (for p) if this progression misses the interval $[Z_p - L(r), Z_p + L(r)]$. By (2), no active step whose two ends lie in the window starts on a cut level.

Lemma 9 (cut levels). *Let $\delta > 0$, let $t \in \mathbb{R}$ with $t/\delta \notin \mathbb{Q}$, and let $0 \leq 2L < \delta$. There is N such that for every $Z \in \mathbb{R}$ some $q \in [1, N]$ and some $q \in [-N, -1]$ satisfy $(tq + \delta\mathbb{Z}) \cap [Z - L, Z + L] = \emptyset$.*

Proof. Divide by δ : with $\theta = t/\delta$ and $\eta = L/\delta < \frac{1}{2}$ the point $q\theta - Z/\delta$ of $\mathbb{R}/\mathbb{Z}$ must enter an arc of length $1 - 2\eta > 0$. By Dirichlet's approximation theorem there are $k \geq 1$ and $j \in \mathbb{Z}$ with $0 < |k\theta - j| < 1 - 2\eta$ ($\neq 0$ as $\theta \notin \mathbb{Q}$). Stepping q by k moves the point by the fixed small amount $k\theta - j$, which cannot jump over the arc; starting at $q = \pm 1$, it enters the arc within $\lceil 1/|k\theta - j| \rceil + 1$ steps. $\square$

We use it with $t = -5$: consecutive rows of teeth are shifted by $5 \equiv \delta/\varphi \pmod{\delta}$, and $5/\delta = 2 + \varphi$ is irrational — a golden rotation.

4.5 Confinement

Proposition 10. *For every $r < r_*$ there is $M(r)$ such that every orbit of $\operatorname{GG}_5(r)$ has at most $M(r)$ points. Hence $\operatorname{GG}_5(r)$ is finite.*

Proof. Besides ψ use the rotated functional $\psi_2(v) = \psi(\zeta^{-1}v) = 2\langle \zeta u, v \rangle$, with level $\ell_2(v) = \ell(\zeta^{-1}v)$; since ζ^{-1} preserves μ_5 , classes and window distances, everything above holds for it, with its own centre Z'_p . As $2L(r) < \delta$, Lemma 9 gives N , depending on r only, and cut levels $q_- < 0 < q_+$ for (ψ, Z_p) and $q'_- < 0 < q'_+$ for (ψ_2, Z'_p) , all in $[-N, N]$. Let

$$V = \{v \in W_p : \operatorname{cl}(v) \in \{0, 1\}, 5q_- < \ell(v) \leq 5q_+, 5q'_- < \ell_2(v) \leq 5q'_+\}.$$

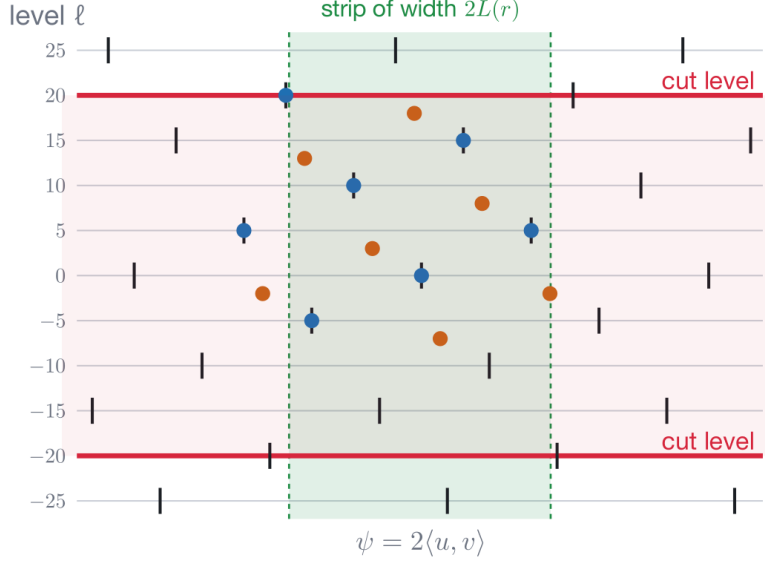


Figure 6. Levels of the orbit of Fig. 5 ($r = 2$). Vertical: the level ℓ ; horizontal: ψ . On each row $5q$ the teeth mark the progression $-5q + \delta\mathbb{Z}$; the green strip is $[Z_p - L(r), Z_p + L(r)]$. Rows whose teeth all miss the strip are cut levels (red): no active step within the window leaves them upwards, so the pivots (blue) and tips (orange) that lie in the window stay in the band between the two nearest cut levels.

V is closed. A typed step from $v \in V$ into the window lands in class 0 or 1. It could leave the first band only by an active step from a pivot on level $5q_+$ or onto a pivot on level $5q_-$, with both ends in W_p ; but $q_{\pm}$ are cut levels. The same holds for ℓ_2 . The dumbbell $(0, 1)$ has levels $(0, 3)$ and $(0, -2)$, inside both bands, and one of its ends lies in W_p because p lies in a disk; so Lemma 6 applies.

V is small. For $v \in V$ we have $|v| \leq 1 + r$, hence $|\psi(v)|, |\psi_2(v)| \leq 2 + 2r$, and $|\ell(v)|, |\ell_2(v)| \leq 5N$; by Lemma 7(i), $\delta|B(v)| \leq |\psi(v)| + |\ell(v)|$, and likewise $\delta|B_2(v)| \leq |\psi_2(v)| + |\ell_2(v)|$ for the golden part $B_2(v) = B(\zeta^{-1}v)$ of $\psi_2(v)$. The four integers ℓ, B, ℓ_2, B_2 determine v : in a basis of Λ they are given by an integer matrix of determinant -5 . So V lies in a box of lattice points whose size depends on r only. By Lemma 6 the pivot, or the tip, of every point of $O_r(p)$ lies in V ; a tip differs from its pivot by a unit, whose coordinates lie in $\{-1, 0, 1\}$; so the pivot lies in the box enlarged by 1 in each coordinate, and $O_r(p)$ has at most 5 times as many points as this enlarged box. The group is finite by Proposition 3. $\square$

Propositions 5 and 10 are parts (a) and (b) of Theorem 1.

4.6 Why the same number twice

The two halves meet at the same radius for visible reasons (Fig. 7). The words of Lemma 4 act on the whole chord only if $E'E$ lies in the lens, that is $|E + 1| \leq r$; the cut needs the strip, of width $2L(r)$, to be narrower than the period δ . Both conditions switch at $r^2 = 3 + \varphi$. In both halves the decisive points are the tips of the lens: the maximum in Lemma 8 is attained there, and at $r = r_*$ the tips are the lattice points $\pm(\zeta - \zeta^{-1})$, where the three words touch the rims.

5 The formal proof

Theorem 1 is proved in Lean 4 [2] (version 4.34.0) with Mathlib [3] (commit 5ed2965) by the library CriticalRadiusFive, about 2 000 lines in eleven modules (Table 1). The library, the files

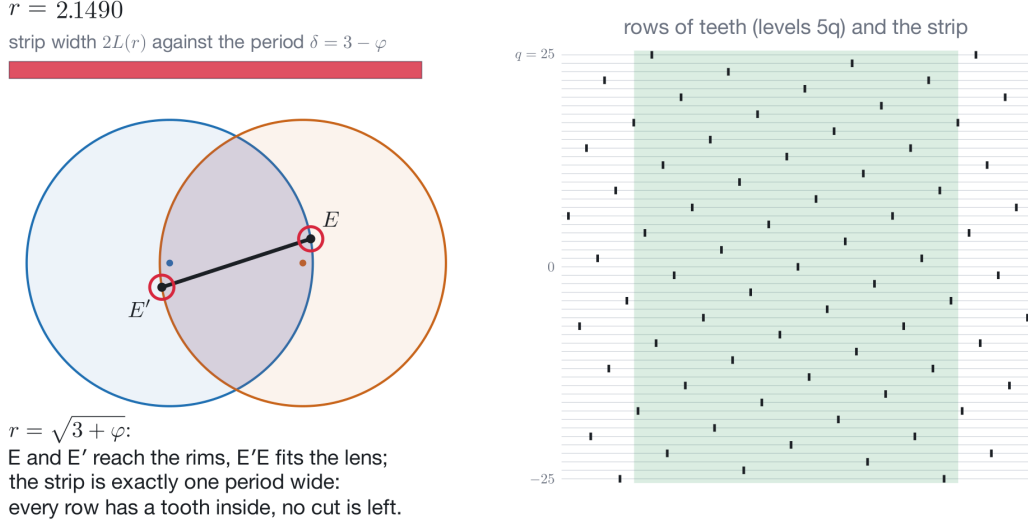


Figure 7. At $r = r_*$: the chord $E'E$ fits the lens exactly (left), and the strip, of width $2L(r)$, is exactly one period wide, so every row of teeth meets it and no cut level is left (right, rows $q = -25, \dots, 25$). For $r < r_*$ the strip is narrower and cut levels appear on both sides.

named below and the animations are at <https://github.com/eric-vergo/critical-radius-five> (tag v1.0); a blueprint linking every statement to its declaration is at <https://eric-vergo.github.io/critical-radius-five/>. The formal statements are those quoted in Section 2; their proofs use the axioms `propext`, `Classical.choice` and `Quot.sound` and nothing else, as the file `Test.lean` checks with `#print axioms`.

Module	Content
Defs	the turns, $GG_n(r)$, the critical radius (general n)
Basic	how the turns act; orbits of points off both disks
Finiteness	Propositions 2 and 3 ; the critical radius as a threshold
Words	words acting on segments; convexity (proof of Lemma 4)
Cyclotomic	exact arithmetic in $\mathbb{Z}[\zeta]$, classes modulo $1 - \zeta$, the norm
CircleRotation	irrational rotations: no return, and Lemma 9
UpperBound	Lemma 4 , Proposition 5
Lens	Lemma 8
PivotWalk	the normal form (1) and Lemma 6
LowerBound	Lemma 7 , (2), Proposition 10
Main	Theorem 1

Table 1. The modules of the formal proof.

What exactly is checked. The file `Challenge.lean` imports only `Mathlib`. It contains the five definitions of `Defs` (the turn of a disk, the two generators, the group and the critical radius) and the two theorems of Section 2 with `sorry` in place of proofs; a reader who wants to know what has been proved needs to read only this file. The file `Solution.lean` states the same two theorems and proves them from the library. The comparator [4], configured by `comparator.json`, checks that each of the two theorems is proved in `Solution.lean` with exactly the statement it has in `Challenge.lean` — every constant the statement mentions, definitions included, must be identical in the two modules — that the proofs use only the three axioms above, and that both Lean’s kernel and the independent kernel `nanoda` [6] accept the solution. On Linux it runs the builds, the exports (by `lean4export` [5]) and the `nanoda` replay inside the `landrun`

sandbox [7]; the Lean kernel replays the exported solution in the comparator’s own process. On a fresh clone at tag v1.0 (macOS, unsandboxed run with the documented opt-in; comparator d03acab, lean4export 076e8e5, nanoda 68d5ca9) the comparator reported “nanoda kernel accepts the solution”, “Lean default kernel accepts the solution” and “Your solution is okay!”. The sandboxed Linux run, the comparator job of the repository’s continuous integration, passed on commit 39b3e01.

Scope. Every numbered statement of Sections 2 to 4 and equations (1) and (2) are formalised (Appendix A); the uniform orbit bound of Theorem 1(b) is proved inside `finite_GG_five_of_lt`, which needs no separate case $r < 1$. Not formalised, because nothing depends on them: the isomorphism $GG_5(r) \cong C_5 \times C_5$ for $0 < r < 1$; the remark in Section 3 and the length ratio $1 : \varphi$ of the pieces; the density of Λ ; that p is a prime of norm 5 (the formal proof uses only the class map `cl`); the “only if” of $2L(r) < \delta \iff r^2 < 3 + \varphi$; the value -5 of the determinant in the proof of Proposition 10 (the formal proof bounds the coordinates of v directly, by linear integer arithmetic); and the discussion in the last subsection of Section 4. The figures were computed in floating point and illustrate; they prove nothing.

Reproducing. `lake exe cache get` && `lake build` builds the library, the challenge, the solution, the axiom audit and the reading guide `Check.lean`; `scripts/verify-comparator.sh` runs the comparator. The figures are the final frames of the animations in `animations/`, rendered by `animations/render.sh`.

A From the paper to the Lean declarations

All names live in the namespace `CriticalRadiusFive`; `Cyc` is Λ with coordinates in the basis $1, \zeta, \zeta^2, \zeta^3$.

Paper	Lean
$a, b, GG_5(r), r_c(5)$	<code>turn, genA, genB, GG, criticalRadius</code>
Proposition 2	<code>orbit_subset_orbit_of_le</code>
Proposition 3	<code>finite_of_encard_orbit_le</code>
consequences in Section 2	<code>finite_GG_of_le, infinite_of_infinite_orbit,</code> <code>finite_GG_iff_of_forall_lt, criticalRadius_eq_of_forall_lt</code>
words on segments, convexity	<code>wordPerm, admissible_of_mem_segment, wordPerm_apply_of_mem_segment</code>
E ; exact disk tests	<code>chordE; inDisk, admissibleZ, movesZ</code>
Lemma 4	<code>piece1, piece2, piece3</code>
the golden rotation of the chord	<code>chordRot, exists_mem_GG_apply_smul_chordE</code>
no return of an irrational rotation	<code>iterate_injective_of_irrational,</code> <code>infinite_orbit_of_irrational_rotation</code>
Proposition 5	<code>infinite_orbit_zero, infinite_GG_five</code>
Λ , classes, units	<code>Cyc, Cyc.cls, Cyc.units, Cyc.cls_of_mem_units</code>
window, closed sets	<code>Window, EdgeClosed</code>
(1), Lemma 6	<code>Good, reach, orbit_subset_reach</code>
u, ψ, ℓ	<code>cutU, psi, psi_eq, level</code>
Lemma 7 (i), (ii), (iii)	<code>psi_add_level; level_of_mem_units, psi_of_level_eq_three;</code> <code>level_emod_five</code>
Lemma 8	<code>abs_re_mul_add_le (with lens_linear_le)</code>
(2); $2L(r) < \delta$	<code>abs_psi_sub_cutCenter_le; two_mul_lensWidth_lt</code>
Lemma 9; $5/\delta \notin \mathbb{Q}$	<code>exists_cut_levels; irrational_neg_five_div</code>
cut levels of ψ	<code>IsCutLevel, exists_isCutLevel</code>
no step crosses a cut level	<code>level_ne_of_isCutLevel, level_mem_band</code>

Paper	Lean
the set V ; V is closed	<code>cutSet; edgeClosed_cutSet, good_cutSet</code>
V is small	<code>abs_psiGold_le, mem_box_of_abs_le, box, mem_box_of_mem_cutSet</code>
Proposition 10	<code>finite_GG_five_of_lt</code>
Theorem 1	<code>finite_GG_five_iff, criticalRadius_five</code>

References

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